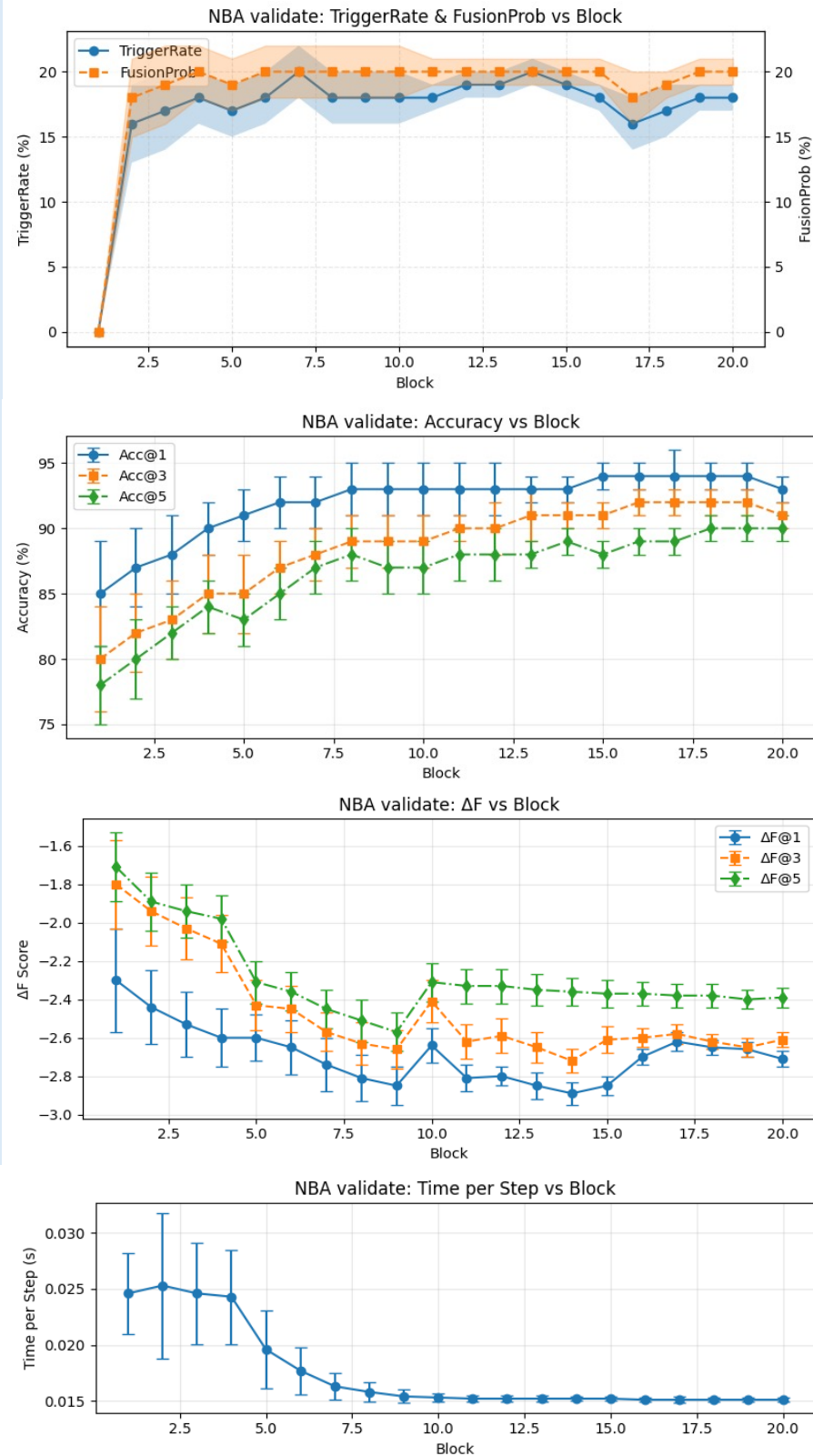
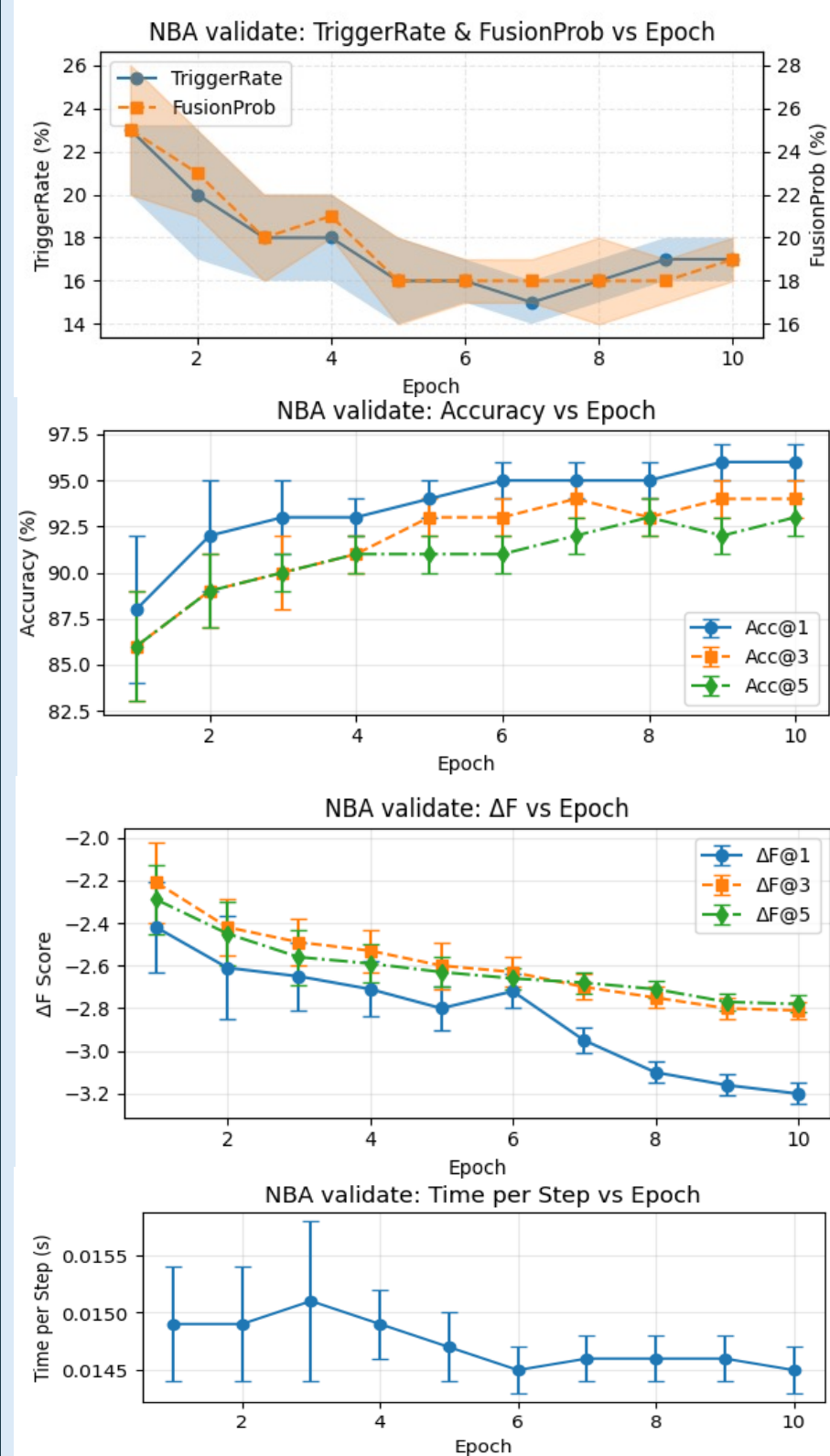


Validation during Pretraining



Validation during Full Training



Top-5 Rules in Pretraining (Blockwise Wake-Sleep)

- $\text{Close}(D1, A) \wedge \text{ClearFront}(A) \rightarrow \text{Dribble}(A)$
confident = 0.75, weight = 0.82
- $\text{Close}(D1, A) \wedge \text{ClearFront}(A) \rightarrow \text{Shoot}(A)$
confident = 0.63, weight = 0.28
- $\text{FrontClose}(D1, A) \wedge \text{RightClose}(D2, A) \wedge \text{LeftOpen}(A) \rightarrow \text{Dribble}(A)$
confident = 0.57, weight = 0.27
- $\text{FrontClose}(D1, A) \wedge \text{RightClose}(D2, A) \wedge \text{LeftOpen}(A) \rightarrow \text{Pass}(A, P1)$
confident = 0.68, weight = 0.73
- $\text{Far}(D1, A) \wedge \text{RimClose}(A) \wedge \text{GoodAngle}(A) \rightarrow \text{Shoot}(A)$ confident = 0.88, weight = 1

Top-5 Rules in Full Training (Full Wake-Sleep)

- $\text{Close}(D1, A) \wedge \text{ClearFront}(A) \rightarrow \text{Dribble}(A)$
confident = 0.88, weight = 0.95
- $\text{Close}(D1, A) \wedge \text{ClearFront}(A) \rightarrow \text{Shoot}(A)$
confident = 0.73, weight = 0.05
- $\text{FrontClose}(D1, A) \wedge \text{RightClose}(D2, A) \wedge \text{LeftOpen}(A) \rightarrow \text{Pass}(A, P1)$
confident = 0.89, weight = 1
- $\text{Far}(D1, A) \wedge \text{RimClose}(A) \wedge \text{GoodAngle}(A) \rightarrow \text{Shoot}(A)$ confident = 0.92, weight = 1
- $\text{BackdoorCut}(P1) \wedge \text{DefenderBeaten}(D1, P1) \rightarrow \text{Pass}(A, P1)$ confident = 0.86, weight = 0.75